

REVIEW

Impact of High Dietary Salt on Neurovascular and Cognitive Functions

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ABSTRACT: Dietary sodium intake in most world regions substantially exceeds physiological requirements, with well-documented consequences for cardiovascular and cerebrovascular health. Although the contribution of dietary salt to blood pressure elevation is well established, evidence from both experimental and human studies indicates that excess sodium adversely affects the cerebral vasculature and cognitive function through mechanisms that operate independently of blood pressure. Here, we review this evidence, integrating findings from animal models with observational and clinical data from human studies. Rodent studies consistently demonstrate that high-salt diet impairs cerebrovascular function and promotes cognitive decline, through mechanisms including IL (interleukin)-17A-mediated endothelial nitric oxide dysfunction, tau hyperphosphorylation, and neuroinflammation. In humans, growing evidence indicates that high dietary sodium accelerates cognitive decline and cerebral small vessel disease progression, though methodological limitations preclude definitive conclusions about causality in most studies. Critically, emerging evidence suggests that the relationship between dietary sodium and cognitive aging is nonlinear, with very low intake also associated with adverse outcomes. Taken together, the available evidence supports a role for dietary sodium in cerebrovascular and cognitive dysfunction that extends beyond its pressor effects, though adequately powered longitudinal studies combining gold-standard assessment of sodium intake with neuroimaging and cognitive end points are needed to establish causality with greater confidence.

Key Words: blood pressure ■ cognitive dysfunction ■ neuroimaging ■ nitric oxide ■ risk factor

Sodium chloride consumption across most of the world substantially exceeds physiological requirements, with well-documented consequences for cardiovascular and cerebrovascular health. A substantial body of epidemiological evidence links elevated dietary sodium to increased risk of stroke, cerebral small vessel disease, and cognitive decline, effects that persist after adjustment for blood pressure (BP). Although hypertension remains the dominant modifiable risk factor for cerebrovascular diseases, and the contribution of dietary salt to BP elevation is well established, the adverse effects of excess sodium on the brain extend beyond its pressor effects. Emerging evidence indicates that dietary salt can trigger immune-mediated pathways that impair cerebrovascular function, and that these mechanisms, operating independently of, or in concert with, BP-mediated injury, may together account for the cognitive consequences of high sodium intake. This review examines the evidence of cerebrovascular and cognitive dysfunction associated with excess dietary sodium, integrating experimental

findings from animal models with observational and clinical data from human studies.

Neurovascular and Cognitive Effects of HSD: Evidence From Animal Models

Besides its effect on peripheral circulation,¹ multiple studies have established that high-salt diet (HSD) exerts profound effects on the cerebral vasculature through mechanisms that are independent from BP elevation.^{2–7} Furthermore, there is increasing evidence that HSD may also impair cognitive function and that cerebrovascular dysfunction may play a role in mediating these effects.^{7,8} Across the studies reviewed here, HSD regimens in rodents typically consisted of chow containing 4% to 8% NaCl (roughly 5–20× the 0.4%–0.8% NaCl content of standard rodent chow), often combined with saline drinking water, for durations ranging from days to several weeks. These intakes substantially exceed typical human high-sodium exposure.¹ Unless otherwise

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Nonstandard Abbreviations and Acronyms

A1R	adenosine A1 receptor
BP	blood pressure
CBF	cerebral blood flow
Cdk5	cyclin-dependent kinase 5
ENLIGHTEN	Exercise and Nutritional Interventions for Neurocognitive Health Enhancement
eNOS	endothelial NO synthase
HSD	high-salt diet
IL-17RA	IL-17 receptor A
Lcn2	lipocalin-2
MMSE	Mini-Mental State Examination
nNOS	neuronal NO synthase
NO	nitric oxide
PP2A	protein phosphatase 2A
ROCK	Rho-kinase
WHIMS	Women's Health Initiative Memory Study

specified, rat studies used Sprague-Dawley rats and mouse studies used C57BL/6 mice, with HSD initiated at ≈2 to 4 months of age. Blood flow assessments in the studies reviewed were performed under anesthesia with artificial ventilation and femoral artery cannulation for blood gas/BP monitoring. Mouse studies used urethane/α-chloralose, the gold-standard regimen for cerebrovascular reactivity studies,⁹ while rat studies used pentobarbital. Irrespective of the anesthetic used, studies in rodents have consistently revealed that HSD selectively suppresses endothelium-dependent vasodilation without altering smooth muscle responsiveness. In rat pial arterioles, HSD blunted the dilatory response to acetylcholine, a response dependent on endothelial nitric oxide (NO), while leaving the response to NO donors fully intact, pointing to a defect at the level of endothelial NO production.² Identical effects were observed in isolated middle cerebral arteries and basilar arteries, where acetylcholine-induced vasodilation was markedly attenuated but response of vascular smooth muscle cells to NO was preserved.⁴ Bradykinin-induced dilation, which relies on COX-1 metabolites rather than NO,¹⁰ remained unaffected.⁴ Direct DAF-2 fluorescence measurements confirmed reduced NO availability in cerebral vessel walls from HSD animals.⁴ Yet, in contrast to findings in peripheral vascular beds, this impairment was not attributable to NO scavenging by reactive oxygen species,⁴ as reactive oxygen species scavengers failed to rescue cerebral endothelial function. Complementary evidence comes from studies of flow-induced dilation, the vasodilatory response to increased intraluminal pressure, which was similarly reduced in isolated middle cerebral arteries

from rats fed a HSD for as little as 1 week.¹¹ In contrast to the reactive oxygen species-independent mechanism identified in other cerebrovascular preparations, flow-induced dilation impairment in this model was rescued by the superoxide dismutase mimetic TEMPOL, implicating oxidative stress as the underlying mechanism. Reduced expression of glutathione peroxidase 4, together with elevated superoxide production in peripheral leukocytes, suggested that impaired antioxidant capacity and reduced NO bioavailability contribute to impair cerebrovascular dilation.¹¹

Beyond the reduced NO availability, middle cerebral arteries from HSD rats exhibited paradoxical vasoconstriction and vascular smooth muscle depolarization in response to both hypoxia and the stable prostacyclin analog iloprost, indicating a broader disruption of vasodilator signaling in cerebral arteries.³ Notably, these impairments were reversible on returning to a low-salt diet, although recovery was surprisingly slow, requiring at least 2 weeks regardless of the duration of prior salt exposure.¹² HSD has also been shown to disrupt cerebral autoregulation,¹³ the intrinsic capacity of the cerebral vasculature to maintain stable blood flow across a range of perfusion pressures. Both short-term and chronic HSD impaired autoregulation in rats, as evidenced by progressive reductions in laser Doppler flux during graded hemorrhage-induced hypotension.¹³ These findings extend the cerebrovascular consequences of HSD beyond vasodilator signaling to a broader impairment of pressure-flow regulation, with potential implications for cerebral vulnerability across the pressure spectrum, both during hypotensive episodes and in the context of hypertension.

Discordant findings come from a study in APP/PS1 mice, a transgenic model of amyloid-β pathology that develops progressive plaque deposition and is widely used to study Alzheimer disease. In this model, a HSD produced results that contrast markedly with the broader literature.¹⁴ Rather than reducing cerebral blood flow (CBF), HSD increased cerebral perfusion in both APP/PS1 mice and control littermates.¹⁴ A subsequent study from the same group, however, found that HSD reduced vascular density in hippocampal and cortical areas of APP/PS1 mice, without a corresponding increase in perfusion, while wild-type controls did show increased CBF.¹⁵ These findings are surprising given the overwhelming and consistent evidence that HSD impairs cerebrovascular function and reduces CBF in wild-type animals and suggest that the presence of Alzheimer disease related pathology may fundamentally alter the cerebrovascular response to dietary sodium excess, though the mechanisms underlying these effects remain unclear.

The molecular basis of HSD-induced endothelial dysfunction has been further characterized in mice. HSD reduced resting CBF in both cortex and hippocampus, as revealed by arterial spin labeling magnetic resonance

imaging.⁷ The CBF response to acetylcholine was similarly diminished, and DAF-based measurements confirmed lower basal NO in isolated cerebral arteries of HSD mice.⁷ Critically, the hyperemic response to whisker stimulation was preserved, establishing that HSD targets endothelial NO generation without disrupting neurovascular coupling.⁷ Supplementing drinking water with L-arginine normalized both endothelial vasodilation and vascular NO levels, and all deficits resolved on return to a normal sodium diet, confirming reversibility of the HSD effects. HSD selectively increased eNOS (endothelial NO synthase) phosphorylation at the inhibitory Thr⁴⁹⁵ site in isolated cerebral arteries without altering the activating Ser¹¹⁷⁷ site,⁷ pointing to suppression of eNOS activity as the mechanism of impaired NO-mediated vasodilation induced by HSD. This effect was downstream of ROCK (Rho-kinase), since the ROCK inhibitor Y27632 restored endothelial function.⁷ An important conceptual advance is the evidence that Thr⁴⁹⁵ eNOS phosphorylation and downstream ROCK activation appear to be driven by IL-17A, a proinflammatory cytokine whose circulating levels are raised by HSD through expansion of gut Th17 cells.^{7,16,17} Th17 cells are specialized producers of IL-17A, a cytokine that plays important roles in mucosal defense but exerts potent proinflammatory effects on the vasculature when chronically elevated.¹ Genetic deletion of IL-17A or anti-IL-17A antibody administration fully prevented the cerebrovascular deficits of HSD, while recombinant IL-17A administration in mice fed a normal diet reproduced them, establishing IL-17A as both necessary and sufficient.⁷ In cultured brain endothelial cells, IL-17A

drove inhibitory Thr⁴⁹⁵ eNOS phosphorylation through ROCK, and Y27632 blocked this effect, directly linking the immune signal to the molecular mechanism identified in HSD animals. Similar findings were subsequently reported in DOCA-salt hypertension, where elevation of circulating IL-17A derived from gut Th17 cells signals through IL-17RA (IL-17 receptor A) on cerebral endothelial cells to increase eNOS phosphorylation on Thr⁴⁹⁵, impairing endothelial NO production and NO-dependent vasodilation.¹⁸

These cerebrovascular deficits were paralleled by cognitive impairment, and the same IL-17A blockade that rescued CBF regulation also restored cognitive performance, while recombinant IL-17A in normally fed mice reproduced both deficits.⁷ The data provided the first direct evidence that cerebrovascular dysfunction drives cognitive impairment in HSD models. A subsequent study uncovered the molecular link between the endothelial NO deficit and cognitive impairment.⁸ When endothelial NO is reduced by HSD, calpain in neurons becomes de-nitrosylated and constitutively active, cleaving p35 to p25 and thereby overactivating Cdk5 (cyclin-dependent kinase 5)⁸ (Figure). This drives phosphorylation of tau at Ser²⁰²/Thr²⁰⁵ and Thr²³¹ and increases insoluble tau in brain homogenates.⁸ Tau hyperphosphorylation and cognitive impairment were also observed in female mice, indicating that this effect is not sex-restricted.⁸ Tau-null mice and anti-tau antibody-treated animals were each protected from cognitive impairment despite persistent CBF reduction, establishing tau accumulation as the final mediator of the cognitive deficit.⁸ Cdk5 inhibition

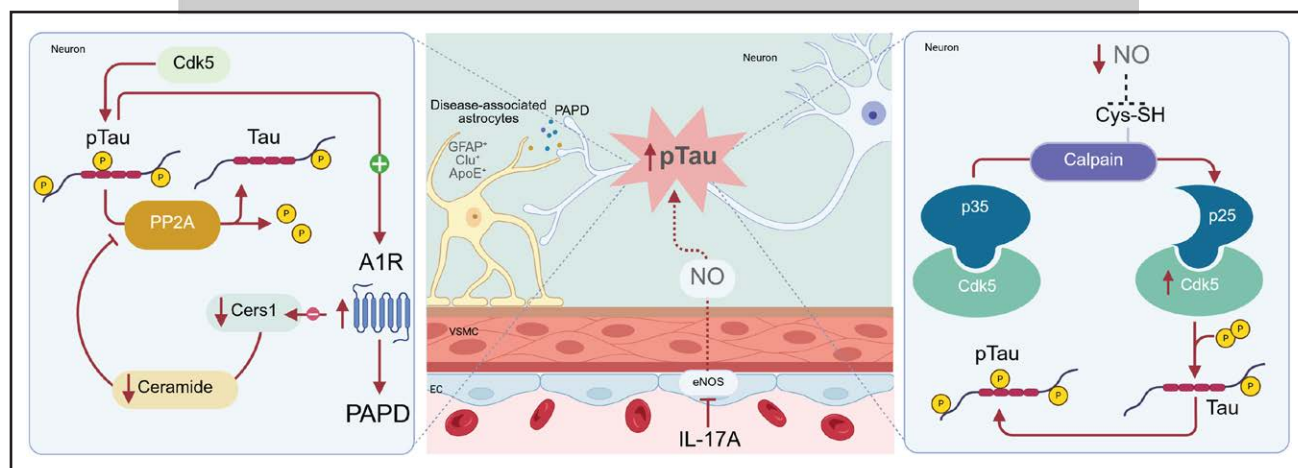


Figure. Mechanisms linking high-salt diet (HSD)-induced endothelial nitric oxide (NO) deficiency to neuronal tau hyperphosphorylation and astrocyte-mediated neuroinflammation.

HSD elevates circulating IL (interleukin)-17A, which suppresses eNOS (endothelial nitric oxide synthase) activity in cerebral endothelial cells (EC), reducing NO bioavailability. **Right panel:** Reduced eNOS-derived NO reduces S-nitrosylation of neuronal calpain (Cys-SH), rendering it constitutively active. Active calpain cleaves the CDK5 activator p35 to its truncated form p25, which drives CDK5 hyperactivation and tau hyperphosphorylation (pTau). **Left:** Hyperphosphorylated tau upregulates A1R (adenosine A1 receptor) signaling, which suppresses Cers1-dependent ceramide synthesis and thereby inhibits PP2A (protein phosphatase 2A), further enhancing tau phosphorylation in a self-amplifying cycle. CDK5 activation further contributes to this feedforward loop. Tau-burdened neurons secrete proteins that promote disease-associated astrocyte-like reactivity (PAPD). **Middle,** PAPD, principally Lipocalin-2 (Lcn2), recruits disease-associated astrocytes (GFAP⁺Clu⁺ApoE⁺) in the hippocampus, amplifying neuroinflammation and synaptic dysfunction. Together, these interconnected pathways link dietary salt excess to tau pathology and cognitive impairment through vascular, neuronal, and glial mechanisms. VSMC indicates vascular smooth muscle cells.

with TFP5, a peptide inhibitor that selectively targets the Cdk5/p25 interaction, reduced tau phosphorylation and prevented behavioral deficits.⁸ Importantly, nitrosylation was markedly suppressed in eNOS-null mice, but not in nNOS (neuronal NO synthase)-null mice, strongly indicating the essential role of eNOS-derived NO in regulating nitrosylation and activity of neuronal calpain.⁸ These findings were the first to define a complete pathological cascade from an endothelial defect to cognitive failure through a neuronally specific mechanism.

An important observation emerging from these studies is the dissociation between CBF reduction and cognitive impairment: tau-null mice and anti-tau antibody-treated animals remained cognitively intact despite persistent CBF reduction, indicating that hypoperfusion alone is not sufficient to cause cognitive failure and that tau accumulation is the required link between vascular deficit and cognitive dysfunction. Further evidence for this dissociation came from a more recent study examining the vascular mechanisms underlying HSD-induced CBF reduction.¹⁹

Capillary stalls, arrest of blood flow by leukocytes or red blood cells in individual capillary segments, have emerged as a significant cause of CBF reduction across multiple disease models, including amyloid accumulation, stroke, diabetes, and seizures.^{20–28} Two-photon in vivo imaging revealed that HSD substantially increases neutrophil stalling in cortical capillaries, detectable as early as 4 weeks and persisting up to 26 weeks.¹⁹ Depletion of circulating neutrophils with anti-Ly6G antibodies reduced stalled capillaries to near-normal levels and restored red blood cell speed and volumetric flow, yet left tau phosphorylation and cognitive performance entirely unaltered,¹⁹ reinforcing the dissociation between hemodynamics and cognition, and confirming that the cognitive decline associated with HSD is independent of blood flow reduction.

Complementary evidence for a HSD-tau link comes from a metabolomic study in aged mice (12 months-old),²⁹ in which 3 months of HSD disrupted both glycolysis and the tricarboxylic acid cycle, reducing the expression of key enzymes including citrate synthase and malate dehydrogenase 2 in the hippocampus and cortex, and inducing tau hyperphosphorylation at Thr¹⁸¹, Thr²⁰⁵, and Thr²³¹ without increasing total tau levels, alongside synaptic protein loss and spatial memory deficits in the Morris water maze.²⁹ These findings suggest that HSD-induced energy metabolism disruption may represent an additional, vascular-independent pathway through which dietary salt promotes tau pathology and cognitive impairment, though mechanistic conclusions from the accompanying cell culture experiments are limited by the use of supraphysiological NaCl concentrations.

The relationship between HSD and amyloid pathology has also been examined. One study reported elevated circulating amyloid- β levels in aged rats (22–24 months

old) on HSD,³⁰ although the significance of this finding is unclear given that serum amyloid- β reflects peripheral production and clearance as much as brain pathology. Consistent with this, we found that HSD did not affect soluble or insoluble amyloid- β levels in the brain,⁸ further suggesting that the cognitive impairment observed in HSD models is not driven by amyloid but rather by tau accumulation.⁸

Building on these findings, recent work has revealed that HSD-induced tau pathology may trigger a broader neuroinflammatory response mediated by astrocytes.³¹ Rui et al demonstrated that HSD induces spatial memory impairment in mice, together with evidence of impaired synaptic transmission.³¹ HSD led to prominent reactivity of disease-associated astrocytes in the hippocampal CA3 region, an effect absent in tau-knockout mice fed HSD and reproduced by direct induction of tau pathology in CA3 neurons, establishing tau as the necessary driver.³¹ The mechanism involves a self-amplifying cycle: neuronal tau upregulates A1R (adenosine A1 receptor) signaling, which suppresses the Cers1-ceramide-PP2A (protein phosphatase 2A) axis and thereby further enhances tau phosphorylation (Figure). Consistent with previous evidence,⁸ Cdk5 activation was also observed in this study, suggesting that both phosphatase inhibition and kinase activation converge to drive tau hyperphosphorylation. In parallel, tau-burdened neurons secrete a set of proteins, collectively termed proteins associated with promoting disease-associated astrocyte-like astrocyte reactivity, that recruit disease-associated astrocytes (Figure); among these, Lcn2 (lipocalin-2) was identified as the principal mediator, with knockdown of neuronal Lcn2 or its astrocytic receptor 24p3R substantially attenuating disease-associated astrocytes reactivity and neuroinflammation.³¹ These findings reveal that the endothelial NO deficit initiated by HSD sets off a cascade in which tau hyperphosphorylation, glial reactive transformation, and sustained cytokine release mutually reinforce, deepening the synaptic and cognitive impairment associated with chronic dietary salt excess.

In apparent contrast to these findings, HSD has also been shown to remodel glial networks in hypothalamic circuits through a mechanism involving accumulation of reactive microglia around vasopressin-secreting neurons, and phagocytosis of astrocytic processes, leading to impaired glutamate clearance and increasing tonic activation of extrasynaptic NMDA receptors and contributing to salt-dependent hypertension.³² Whether these divergent glial responses reflect differences in brain region, cell type, or HSD regimen remains to be determined, but together they illustrate that HSD may engage multiple, region-specific glial remodeling programs with distinct functional consequences.

A further study provided additional evidence that HSD drives neuroinflammation, demonstrating that, in female mice, HSD feeding elevated cortical IL-1 β , IL-6, and

TNF- α , increased levels of the microglial marker Iba1, disrupted BBB integrity via reduced claudin-5 expression, and induced cortical and hippocampal apoptosis.³³ However, the mechanisms through which HSD may activate microglia were not directly investigated, and the characterization of microglial responses to HSD was superficial, with future studies needed to determine whether HSD drives specific microglial states, morphological changes, or phagocytic dysfunction. Nevertheless, the authors proposed gut dysbiosis as a potential upstream contributor, with HSD reducing SCFA-producing bacteria and fecal SCFA concentrations, which may in turn compromise BBB integrity and promote neuroinflammation via the gut-brain axis.³³ In contrast, a study in PLP-haSyn mice, a transgenic mouse model of multiple system atrophy, a neurodegenerative disease characterized by α -synuclein accumulation in oligodendrocytes, found no evidence of HSD-induced neuroinflammation, with microglial activation, neuroinflammation-related cytokines and chemokines, T-cell recruitment, and astrogliosis all unaffected by 3 months of HSD exposure, and no effect on motor decline, neurodegeneration, or α -synuclein accumulation.³⁴ Although the reasons for this discrepancy are unclear, 1 possibility is that neuroinflammation in the PLP-haSyn model is already near maximal due to the underlying disease process, leaving little room for further exacerbation by dietary salt.

Consistent with the link between HSD and tau phosphorylation, Yang et al³⁵ reported that a long-term HSD reduces GLP-1R expression and inhibits autophagic flux through activation of the mTOR/p70S6K pathway in 9 months-old mice, accompanied by tau hyperphosphorylation and synaptic protein loss, further supporting the link between HSD and tau pathology through metabolic signaling, though the precise mechanistic relationships between these pathways require further investigation. Further supporting a HSD-tau link, another recent study reported tau hyperphosphorylation at multiple sites alongside synaptic protein loss and cognitive deficits in mice given 2% NaCl drinking water for 12 weeks.³⁶ However, as HS intake also induced significant hypertension in this model, the independent contribution of dietary salt versus elevated BP to tau pathology cannot be disentangled. In Fischer Brown Norway rats, Morris water maze deficits have been observed in association with dendritic spine loss without hypertension, and comparisons between young (2 months old) and aged (20 months old) animals have revealed that cognitive vulnerability to HSD is age-dependent, with older animals showing long-term memory impairment that young adults escape, an effect accompanied by elevated brain oxidative stress,³⁷ a vulnerability further reinforced by more recent data in both young and aged rats across multiple salt regimens.³⁸ The role of oxidative stress has also been implicated in mice where HSD-induced cognitive impairment has been associated with hippocampal superoxide elevation

and reduced antioxidant enzyme activity.³⁹ Ge et al⁴⁰ further demonstrated that 7 weeks of HSD impaired both short-term spatial memory and long-term contextual fear memory in mice, with long-term memory showing deficits already after 4 weeks, suggesting a time-dependent effect. These behavioral impairments were accompanied by reduced hippocampal LTP, increased reactive oxygen species, decreased antioxidant enzyme activity, and synaptic protein loss which correlated negatively with duration of HSD exposure.⁴⁰ Critically, BP was not elevated in this model, confirming that these effects were independent of hypertension.⁴⁰ Additional evidence of the link between HSD and cognitive impairment was provided by a recent study where 6 months of HSD exposure in mice was associated with spatial memory deficits in the Morris water maze and increased anxiety-like behavior in the open field test.⁴¹ Similarly, Fan et al⁴² reported that 12 weeks of HSD selectively impaired spatial memory in 8 months old mice without affecting anxiety or fear memory and identified disruption of SIRT3/PINK1-mediated mitophagy and mitochondrial fission as the underlying mechanism, linking HSD-induced mitochondrial dysfunction to synaptic protein loss and cognitive impairment.

In contrast to mice, when tau phosphorylation was assessed in rats fed a HSD, despite BP elevation, no significant changes were detected, nor were any alterations in hippocampal A β 42 levels.⁴³ Whether the discrepancy with mouse studies reflects differences in species, salt concentration, duration of exposure, or the presence of additional confounders such as hypertension severity remains to be determined. Taken together, the studies reviewed here provide converging evidence that HSD promotes tau hyperphosphorylation and cognitive impairment. Nevertheless, in some of these studies HSD also induces hypertension, making it difficult to attribute tau pathology specifically to elevated sodium rather than to the hemodynamic consequences of BP elevation, and the absence of tau changes in at least 1 rat model underscores that the relationship may be sensitive to species and experimental conditions.

Beyond effects in adult animals, evidence suggests that HSD exposure during pregnancy may have lasting consequences for cognitive function in the offspring.⁴⁴ Adult offspring of dams fed an HSD exclusively during gestation exhibited short- and long-term spatial memory deficits, impaired hippocampal synaptic transmission, reduced LTP at CA3-CA1 synapses, and decreased expression of the synaptic proteins synaptophysin and PSD-95.⁴⁴ These functional deficits were accompanied by a transient reduction in hippocampal astrocyte number during the juvenile period, with compensatory normalization by adulthood, suggesting that a critical window of astrocyte-dependent synaptic maturation may be disrupted by maternal sodium excess. Notably, the cognitive deficits persisted into adulthood despite offspring being returned to a normal diet after weaning, and were

observed in both male and female offspring, indicating a sex-independent and lasting programming effect of gestational HSD on hippocampal function. Taken together with the finding that HSD recruits reactive astrocytes in the adult hippocampus,³¹ these data suggest that astrocytes represent a cellular target of HSD, with reduced abundance during development and reactive transformation in adulthood both converging on synaptic dysfunction and cognitive impairment.

Overall, there is growing evidence that HSD promotes cognitive impairment across different models and different HSD regimes, affecting both male and female mice. Although several mechanisms have been implicated, a direct functional link between HSD-induced cerebrovascular dysfunction and cognitive decline has been established in only a small number of studies, and the relative contribution of vascular, inflammatory, and metabolic pathways to the overall cognitive phenotype remains to be fully elucidated.

Salt and Cognitive Decline: Human Evidence

Direct evidence of HSD-induced cerebrovascular dysfunction in humans is currently lacking, as the noninvasive assessment of cerebral endothelial function remains technically challenging and has not been systematically applied in the context of dietary salt manipulation. The available human evidence is therefore restricted to peripheral vascular beds, where HSD has been consistently shown to impair endothelium-dependent vasodilation independently of BP elevation.¹ With respect to cognitive function, evidence in humans remains limited, with available studies constrained by imprecise dietary assessment, including the failure of food frequency questionnaires to capture salt added during cooking or at the table, by the difficulty of isolating sodium from other correlated dietary variables, by substantial variability across studies in the intake thresholds used to define high versus low sodium, and by the potential for survivor bias and reverse causation in observational cohorts, all of which preclude definitive causal inferences. Nevertheless, the available evidence, taken as a whole, points toward a detrimental effect of high sodium intake on cognitive function, though the relationship is more complex than a simple linear 1, as discussed below.⁴⁵ The following sections review the available human evidence, organized by study design, from cross-sectional observations to prospective and longitudinal cohort data, serum sodium studies, and finally intervention evidence (Table).

Cross-Sectional Studies

Cross-sectional studies have examined the relationship between sodium intake and cognitive performance across diverse populations, yielding mixed findings. An early cross-sectional study of 44 Irish adults who were first-degree relatives of people with Alzheimer disease

found that sodium intake was significantly higher in the 4 subjects with a Mini-Mental State Examination (MMSE) score outside the normal range compared with those with normal scores, though the extremely small number of cognitively impaired participants limits the conclusions that can be drawn from this study.⁶⁹ Elevated 24-hour urinary sodium excretion was associated with worse scores on the MMSE in 119 adults with newly diagnosed hypertension, though a single urine collection, while preferable to dietary recall methods, limits interpretation given the known day-to-day variability in urinary sodium excretion.⁴⁶ Paradoxically, worse MMSE scores and longer Trail Making Test completion times, a measure of executive function, visuomotor tracking and attention, were observed in 925 older adults in the lowest sodium intake quartile, independent of health-related confounders including hypertension, underscoring the possibility that very low sodium intake may itself carry cognitive risk.⁴⁷ Another cross-sectional study using 24-hour urine collections in community-dwelling adults examined the sodium-to-potassium ratio (Na^+/K^+), a composite measure reflecting both high sodium and low potassium intake, and found that a higher urinary Na^+/K^+ was independently associated with up to 3.4-fold greater odds of mild cognitive impairment, with lower potassium excretion also independently protective,⁴⁸ suggesting that the cognitive consequences of dietary sodium may be modulated by concurrent potassium intake. In contrast, the InCHIANTI study of 1271 adults found no association between 24-hour urinary sodium excretion or dietary sodium and MMSE or Trail Making Test performance, though high nitrate intake was associated with lower BP independently of sodium, raising the possibility that the dietary context in which sodium is consumed may modify its effects.⁴⁹

Prospective and Longitudinal Studies

Prospective and longitudinal studies provide stronger temporal evidence than cross-sectional designs, though heterogeneity in follow-up duration, exposure assessment, and cognitive outcomes limits direct comparability.

The NuAge Study, a 5-year prospective longitudinal study of 1262 cognitively normal older adults (mean age, 74 years), found that among those with low levels of physical activity, sodium intake in the lowest tertile (median, 1.8 g/d) was associated with significantly less cognitive decline on the Modified MMSE over 3 years compared with those with mid (median, 2.634 g/d) or high (median, 3.701 g/d) level sodium intake, an association that remained after adjustment for age, sex, education, waist circumference, diabetes, and overall diet quality. No such association was observed in physically active individuals, for whom cognitive trajectories did not differ across sodium intake tertiles.⁵⁰ These findings suggest that physical activity may confer a protective effect on the cerebrovascular system that attenuates the

Table. Summary of Human Studies Examining the Association Between Dietary Sodium Intake and Cognitive Outcomes

Author, y	Study design	Participants, n	Age, y	Na assessment	Cognitive assessment	Main finding	Highest level reported	Effect
Afsar ⁴⁶	Cross-sectional	119 (HTN)	54.2±16.1	24 h urinary Na (single)	MMSE	Higher urinary Na associated with worse MMSE scores.	Not reported	↑
Rush et al ⁴⁷	Cross-sectional	925	74.5±8.7	Dietary recall	VFT, MMSE, TMT	Lowest Na quartile associated with worse MMSE and longer TMT-B.	≈2.6 g/d (sodium intake)	↑
Wang et al ⁴⁸	Cross-sectional	561	52.7±9.3	24 h urinary Na/K ratio	MMSE	Higher urinary Na/K independently associated greater MCI odds; lower K protective.	≈11.7 g/d (salt intake)	↑
McGrattan et al ⁴⁹	Cross-sectional	989	73.5±8.8	24 h urinary Na + dietary recall	MMSE, TMT	No association between urinary Na or dietary Na and cognitive performance.	Mean 2.4 g/d (sodium intake)	=
Fiocco et al ⁵⁰	Prospective	1262	74.2±0.4	Dietary recall	3MS	Low Na protective in sedentary individuals; no effect in physically active individuals.	≈ 3.1–8.1 g/d (sodium intake)	↑*
Haring et al ⁵¹	Prospective	6426 (post-menopausal women)	65–79	FFQ + 24 h urinary Na	3MS	Na intake did not significantly modify dementia risk; hypertension itself was associated with increased risk; High sodium intake may amplify risk in hypertensive women.	>2.3 g/d (sodium intake)	=
Nowak et al ⁵²	Prospective	1194	74±3	FFQ + urinary biomarkers	3MS	No association between Na intake and cognitive decline or brain changes on MRI.	3.3–11.6 g/d (sodium intake)	=
Milte et al ⁵³	Cross-sectional	617	60.2±3.14	Discretionary salt behavior	TICS-m	Men adding salt during cooking performed worse; women adding salt after cooking performed better.	Not reported	↑*
Klinedinst et al ⁵⁴	Prospective	1787	63±7.5	Discretionary salt behavior	FIT	Salt associated with worse fluid intelligence only in those with AD family history or APOE ε4.	Not reported	↑*
Na et al ⁵⁵	Prospective	4213	59 (Median)	24 h dietary recall + household condiment weighing	TICS-m, Self report of memory status and memory change	Higher Na and Na/K intake associated with worse cognitive function.	≈15.1 g/d (sodium intake)	↑
Liu et al ⁵⁶	Prospective	2041	68.5±6.1	7 consecutive days 24 h urine	MMSE, MoCA, DRS, IQCODE	Global cognition declined progressively faster with increasing salt, independently of BP.	>12 g/d (salt intake)	↑
Liu et al ⁵⁷	Prospective	453	67.5±5.4	24 h urine	CSVD progression (MRI)	Excessive salt intake accelerated CSVD progression.	>12 g/d (salt intake)	↑
Makin et al ⁵⁸	Cross-sectional + Systematic Review	264 (Minor Ischemic Stroke)	67.5 (Median)	Dietary salt intake (self-reported)	CSVD markers (MRI)	Patients not reducing salt more likely to have lacunar stroke, lacunes, microbleeds, severe WMH.	Not reported	↑
Ahn et al ⁵⁹	Observational	705 (Mild to Moderate AD)	74.6±7.9	Dietary salt behavior	SNSB, MMSE, GDS-15, CDR, CGA-NPI and S-IADL	Salt restriction associated with slower clinical progression in dementia severity, cognition.	Not reported	↑
Tong et al ⁶⁰	Prospective + postmortem	1520	80.7±7.2	FFQ	CERAD WLR, BNT, SDMT, DSF/DSB	Lowest quintile (≈1.8 g/d) associated with faster decline and higher tau; highest quintile not significant; U-shaped association with arteriosclerosis.	≈2.8 g/d (sodium intake)	↑
Chuwa et al ⁶¹	Prospective	1208	70.9±6.8	Validated FFQ	PACC	Males with higher Na showed faster episodic recall decline; no effect in females.	≈2.3–6.3 g/d (sodium intake)	↑*
Abe et al ⁶²	Longitudinal ecological	151 countries (GBD data)		National-level Na intake data	Dementia prevalence	Higher Na intake positively associated with dementia prevalence in men and women; persisted after BP adjustment.	Not reported (national-level data)	↑
van der Burgh et al ⁶³	Cross-sectional	8028	63.6±9.6	Serum sodium	MMSE, LDST, Stroop Test, WFT, WLT, PPB	Low serum Na associated with worse memory, executive function, attention, psychomotor speed; not with overall cognition or dementia.	Not reported (serum Na, not intake)	↑

(Continued)

Table. Continued

Author, y	Study design	Participants, n	Age, y	Na assessment	Cognitive assessment	Main finding	Highest level reported	Effect
Lee et al ⁶⁴	Cross-sectional	2541	69.6±6.8	Serum sodium	CERAD WLT, AFT, and DSST	Low serum Na associated with worse domain-specific cognitive performance.	Not reported (serum Na, not intake)	↑
Hui et al ⁶⁵	Cross-sectional	403	71.1±7.8	Serum sodium	MoCA	U-shaped relationship; frailty amplified MCI risk at both extremes.	Not reported (serum Na, not intake)	U
Nowak et al ⁶⁶	Prospective	5435	74±6	Serum sodium	3MS, TMT	Both low and high serum Na associated with cognitive impairment.	Not reported (serum Na, not intake)	U
Chen et al ⁶⁷	Cross-sectional	353	70.6±7	Serum sodium	ADAS13, CDRSB, MMSE	Higher serum Na associated with greater amyloid and tau burden, lower hippocampal volume, worse cognition.	Not reported (serum Na, not intake)	↑
Blumenthal et al ⁶⁸	Randomized controlled trial	160 (adults with cognitive impairments with no dementia)	65.4±6.8	Dietary recall (Na reduction within DASH diet)	TMT, Stroop Test, AFT, DSST, DSF/DSB	Na reduction independently associated with improved executive function; aerobic exercise showed significant main effect; DASH diet alone did not.	Not reported	↑

Effect symbols: ↑ denotes higher sodium associated with worse cognition/ greater risk; =, null finding. U, U- or J-shaped relationship; and U/sex, sex-divergent findings. 3MS indicates modified mini-mental state examination (3MS); AD, Alzheimer disease; ADAS13, Alzheimer's Disease Assessment Scale 13; AFT, Animal Fluency Test; BNT, Boston Naming Test; BP, blood pressure; CCVD, cardiovascular and cerebrovascular disease; CDR, clinical dementia rating; CDRSB, clinical dementia rating sum of boxes; CERAD WLT/WLR, Consortium to Establish a Registry for Alzheimer's Disease Word Learning Test/Word List Recall; CGA-NPI, Caregiver-Administered Neuropsychiatric Inventory; CSVD, cerebral small vessel disease; DASH, Dietary Approaches to Stop Hypertension; DRS, dementia rating scale; DSF/DSB, Digit Span Forward/Backward test; DSST, Digit Symbol Substitution Test; FFQ, food frequency questionnaire; FIT, fluid intelligence test; GBD, Global Burden of Disease; GDS-15, Geriatric Depression Scale-15; HTN, hypertension; IQCODE, informant questionnaire on cognitive decline in the elderly; K, potassium; LOST, Letter Digit Substitution Test; MCI, mild cognitive impairment; MMSE, mini-mental state examination; MoCA, Montreal Cognitive Assessment; MRI, magnetic resonance imaging; Na, sodium; PACC, AIBL Preclinical Alzheimer Cognitive Composite; PPB, Purdue Pegboard Test; SDMT, Symbol Digit Modalities Test; S-IADL, Seoul Instrumental ADL; SNSB, Seoul Neuropsychological Screening Battery; SVD, small vessel disease; TICS-m, telephone interview for cognitive status-modified; TMT, trail-making test; VFT, verbal fluency test; WFT, Word Fluency Test; WLT, Word Learning Test; and WMH, white matter hyperintensities.

*Effect present only in subgroup.

cognitive consequences of high sodium intake, and that the impact of dietary sodium on cognition may be contingent on overall lifestyle context.

The WHIMS (Women's Health Initiative Memory Study), a large prospective study of 6426 women with a median follow-up of over 9 years, found that while hypertension itself was associated with an increased risk of mild cognitive impairment or probable dementia, dietary sodium intake did not significantly modify this risk, with formal tests for interaction remaining nonsignificant regardless of whether sodium was assessed by food frequency questionnaire or corrected using 24-hour urinary excretion data in a subsample.⁵¹ However, sodium intake above 2300 mg/d was associated with increased risk in hypertensive women specifically, raising the possibility that high sodium intake may amplify the adverse cognitive effects of hypertension, though this subgroup finding should be interpreted cautiously given the nonsignificant interaction test.⁵¹ The consistency of the null finding across both assessment methods strengthens confidence in this result, though sodium was assessed only at baseline, precluding evaluation of long-term dietary trajectories and potentially misclassifying participants whose intake changed substantially over the 9-year follow-up. Similarly null findings emerged from a 10-year longitudinal study of 1194 older adults, in which neither cognitive decline nor micro- and macrostructural

brain changes on magnetic resonance imaging were associated with dietary sodium intake assessed by both food frequency questionnaire and urinary biomarkers,⁵² though as with WHIMS, sodium was assessed only at baseline, limiting the ability to capture changes in dietary habits over the follow-up period.

Studies using discretionary salt behavior as a proxy for intake rather than objective measures have also yielded mixed results. Milte et al⁵³ found that men who habitually added salt during cooking performed significantly worse on a brief telephone cognitive assessment, independent of age, education, and physical activity. Notably, women who occasionally added salt after cooking showed better cognitive performance than those who never did, an unexpected finding the authors contextualize within the J-shaped sodium-cognition relationship, as discussed below, wherein very low sodium intake may itself increase the risk of cognitive impairment. These sex-divergent findings add to growing evidence that sodium's cognitive effects are not uniform across sexes. The same proxy measure was employed in a 10-year prospective study of 1787 mid-to-late-aged adults from the UK Biobank, where habitually adding salt to meals was associated with worse fluid intelligence scores, a measure of abstract problem-solving capacity, but only among those with a family history of Alzheimer disease or carrying the apolipoprotein E ε4 allele, suggesting that

genetic susceptibility to Alzheimer disease may amplify the cognitive consequences of high sodium intake.⁵⁴ A methodological strength of this study is that dietary data were collected at 3 time points over the 10-year follow-up, reducing the misclassification error inherent in single-baseline assessments. A more detailed assessment combining 3 consecutive days of 24-hour dietary recall with household condiment weighing was used in a large prospective Chinese cohort study of 4213 adults aged 50 years and above in whom sodium intake reached a median of 8.068 g/d in the highest quartile.⁵⁵ While sodium was not associated with objective cognitive test scores, higher sodium and sodium-to-potassium ratio were associated with self-reported poor and deteriorated memory,⁵⁵ though self-reported measures are generally considered less reliable than objective assessments. This is consistent with the broader observation that food processing, a major driver of excess dietary sodium, substantially reduces the potassium content of certain food categories, contributing to a decline in dietary potassium,⁷⁰ suggesting that the sodium-to-potassium ratio may serve as a more integrative marker of dietary pattern than sodium intake alone. Importantly, the associations of dietary potassium and sodium-to-potassium ratio with cognitive decline were partly mediated by cardiovascular and cerebrovascular disease and by sleep duration, suggesting that salt may exert some of its cognitive effects indirectly through these pathways.⁵⁵

The strongest prospective evidence to date comes from a study that addressed a key limitation of most prior work, reliance on single or short-term dietary assessments, by collecting 7 consecutive days of 24-hour urine samples in both spring and autumn at baseline, capturing both day-to-day and seasonal variation in salt intake, in 2041 older adults aged 60 years and above from Shandong, China.⁵⁶ Cognitive function was assessed with a comprehensive battery including the MMSE, Montreal Cognitive Assessment, Dementia Rating Scale, and the Informant Questionnaire on Cognitive Decline in the Elderly, and participants were monitored over a mean follow-up of 11.4 years.⁵⁶ Global cognitive function declined progressively faster with increasing salt intake, with dementia risk increasing by 75% in the mild intake group (6–9 g/d), 180% in the moderate group (>9–12 g/d) and 330% in the high group (>12 g/d), relative to the low intake reference group (<6 g/d), corresponding to a 59% rise in dementia risk per SD increment in salt intake, independent of BP changes.⁵⁶ Critically, these associations remained significant after adjustment not only for baseline BP but also for visit-to-visit variability in both systolic and diastolic pressure throughout follow-up as well as for smoking, alcohol use, diabetes, dyslipidemia, antihypertensive medication use, and apolipoprotein E genotype, establishing that high salt intake is an independent risk factor for dementia beyond its effects on BP.⁵⁶ Notable limitations include the absence of data

on total caloric, fat, and potassium intake; additionally, the high prevalence of hypertension at baseline (72% of participants) limits the ability to establish that the observed cognitive effects are truly salt-dependent and independent of hypertension, as covariate adjustment for BP, however comprehensive, cannot formally separate the direct effects of salt from those mediated through BP. A companion prospective study from the same group in 423 older adults followed over 5 years demonstrated that excessive salt intake also accelerated the progression of cerebral small vessel disease, with those consuming >12 g/d showing a 7.4-fold increased risk of new white matter hyperintensities and a 6.6-fold increased risk of composite cerebral small vessel disease compared with the low intake group, as well as a significantly increased stroke risk, all independent of BP and baseline white matter hyperintensity volume,⁵⁷ though as in the previously mentioned study, the high prevalence of hypertension at baseline limits the ability to fully exclude a BP-mediated contribution. These findings are further supported by a cross-sectional study in 250 stroke patients by Makin et al,⁵⁸ in which patients who had not reduced their salt intake during adult life were significantly more likely to have lacunar stroke, lacunes, microbleeds, and severe white matter hyperintensities after adjustment for hypertension and age, though salt intake was assessed by self-reported dietary history rather than objective urinary measures, and the cross-sectional design in a selected stroke population limits generalizability.

Beyond cognitively normal populations, the possibility that salt restriction may have disease-modifying potential even after dementia onset is raised by a Korean observational study in patients with mild-to-moderate Alzheimer disease, in which those who habitually added salt or soy sauce to food showed significantly faster progression in dementia severity, cognitive function, and neuropsychiatric symptoms over 3 years of follow-up compared with those who rarely did so.⁵⁹ These findings raise the clinically important question of whether reducing salt intake could represent a modifiable factor in dementia management.

Adding further complexity to this picture, a community-based prospective study of 1520 older adults (mean age, 80.7 years) followed over a mean of 7.6 years with complementary postmortem neuropathological data from 717 participants,⁶⁰ found that compared with the middle quintile (≈ 2.3 g/d), it was the lowest quintile (≈ 1.764 g/d) rather than the highest (≈ 2.8 g/d) that was associated with faster decline in global cognition, episodic memory, and semantic memory, and with higher postmortem tau tangle density among deceased participants.⁶⁰ The highest quintile showed no significant association with cognitive decline or neuropathology, though the highest quintile, while above recommended limits, does not represent the substantially elevated intakes documented in populations where detrimental effects were most pronounced.

The association of low sodium with cognitive decline was attenuated after adjustment for APOE ϵ 4 and disappeared when excluding participants who developed MCI or dementia within 2 years of baseline, raising the possibility that early cognitive changes may have driven dietary modifications rather than low sodium causing cognitive decline. Both the lowest and the fourth quintile were also associated with higher odds of moderate to severe arteriosclerosis,⁶⁰ further suggesting that the relationship between salt and cerebrovascular health is nonlinear. These findings are consistent with an earlier cross-sectional study discussed above, in which lower dietary sodium was associated with worse global cognition and executive function, independently of hypertension, kidney function, and diet quality, with the executive function effect confined to adults aged 80 years and over.⁴⁷

A recent 6-year longitudinal study from the well-characterized Australian Imaging, Biomarkers and Lifestyle cohort examined the relationship between baseline dietary sodium intake and change across 6 cognitive composite domains in 1208 cognitively unimpaired older adults.⁶¹ No association was detected in the cohort as a whole; however, sex-stratified analyses revealed that males with higher sodium intake at baseline showed significantly faster decline in the episodic recall composite over 72 months, an effect that remained after additional adjustment for BP, while no equivalent association was observed in females.⁶¹ The sex-specific finding is noteworthy given that episodic recall is among the earliest and most sensitive cognitive markers of incipient Alzheimer disease, relying heavily on hippocampal circuitry that is particularly vulnerable to cerebrovascular insufficiency. This study is among the methodologically stronger human investigations to date, combining a validated food frequency questionnaire, comprehensive cognitive testing across 6 domains, and a well-characterized longitudinal cohort design. The absence of an effect in the full cohort and in females underlines the importance of sex-stratified analyses in this field.

Extending beyond individual cohorts to the population level, a longitudinal ecological analysis examined national-level sodium intake data from 151 countries alongside Global Burden of Disease dementia prevalence estimates spanning 2010 to 2021.⁶² After adjustment for GDP, aging rates, education, obesity, physical activity, and alcohol intake, higher habitual sodium intake was positively associated with dementia prevalence in both men and women, with the association persisting after adjusting for country-level hypertension prevalence, suggesting a BP-independent contribution.⁶² Although ecological designs cannot establish causation at the individual level, the consistency of this association across such diverse populations and its robustness to adjustment for major macroeconomic and lifestyle confounders provide strong population-level support for the link between dietary sodium and dementia risk.

Serum Sodium and Cognitive Function

Complementing the dietary intake data reviewed above, a body of evidence links circulating sodium levels to cognitive performance, extending the sodium-cognition relationship beyond intake to systemic sodium homeostasis. Two large cross-sectional studies found that low serum sodium was associated with worse performance on domain-specific cognitive tests, particularly memory, executive function, attention, and psychomotor speed, but not with overall cognitive function or incident dementia, reinforcing the view that the relationship between sodium and cognitive health is nonlinear, with risk present at both ends of the sodium spectrum.^{63,64} A cross-sectional study of 403 hospitalized older adults with chronic diseases further demonstrated a U-shaped relationship between serum sodium and MCI risk via restricted cubic spline analysis, with a nadir at ≈ 143 mmol/L, and notably found that both the lowest and highest sodium tertiles were associated with markedly elevated MCI risk specifically in frail individuals, suggesting that frailty amplifies vulnerability to sodium dysregulation at both extremes.⁶⁵ Consistent with this, a prospective cohort study of over 5000 older community-dwelling men found that both low and high serum sodium were independently associated with cognitive impairment and decline, explicitly suggesting a U- or J-shaped relationship.⁶⁶ Furthermore, examining participants spanning the Alzheimer's disease continuum from the ADNI cohort, higher serum sodium was associated with greater amyloid and tau PET burden, lower hippocampal volume, and worse cognitive scores, extending the sodium-cognition relationship to neuropathological end points.⁶⁷ This is consistent with broader data indicating that serum sodium above 142 mmol/L is associated with accelerated biological aging, increased risk of chronic disease, and premature mortality, suggesting that sodium dysregulation carries systemic consequences well beyond the brain.⁷¹

Evidence From Interventional Studies

Evidence linking sodium reduction to cognitive benefit comes from the ENLIGHTEN trial (Exercise and Nutritional Interventions for Neurocognitive Health Enhancement), conducted in sedentary older adults with cognitive impairment without dementia and cardiovascular risk factors, which examined the independent and combined effects of aerobic exercise and the Dietary Approaches to Stop Hypertension diet on executive function over 6 months.⁶⁸ Although aerobic exercise produced significant improvements in executive function and the Dietary Approaches to Stop Hypertension diet alone did not reach significance as a main effect, a posthoc correlational analysis revealed that reduced sodium intake was independently associated with improved executive function, whereas changes in potassium, magnesium, and calcium were not.⁶⁸ This finding must be interpreted cautiously

given its exploratory and correlational nature within a trial not powered to isolate sodium effects, its short 6-month duration precluding any assessment of long-term effects, and the fact that the Dietary Approaches to Stop Hypertension diet encompasses multiple dietary changes beyond sodium reduction. Nevertheless, it represents the only randomized trial data to date suggesting that dietary sodium reduction may contribute independently to cognitive benefit, consistent with the preclinical mechanistic data reviewed above.

CONCLUSIONS

The evidence reviewed here, spanning experimental animal models and human observational and interventional data, converges on the view that dietary salt excess represents a biologically plausible and clinically relevant risk factor for cognitive decline. In rodent models, HSD impairs cerebral endothelial function, reduces cerebral blood flow, promotes tau hyperphosphorylation, triggers neuroinflammation, and disrupts synaptic plasticity through interconnected vascular, immune, and metabolic pathways, several of which operate independently of BP elevation.

Whether these mechanisms operate in humans remains an open question. Direct evidence of HSD-induced cerebrovascular dysfunction in humans is lacking, and the available data on cognitive function are predominantly observational, heterogeneous in design, and methodologically limited. Nevertheless, the balance of evidence from prospective cohort studies, particularly those employing gold-standard urinary biomarkers and long follow-up, points toward a dose-dependent detrimental effect of high sodium intake on cognition, independent of BP. However, this relationship is not linear, and may be shaped by physical activity, genetic susceptibility, sex, and potassium intake, with very low sodium itself potentially carrying cognitive risk.

Several priorities emerge for future research. On the mechanistic side, direct assessment of cerebral endothelial function and neurovascular coupling in humans undergoing controlled dietary sodium manipulation is needed to establish whether the vascular pathways identified in animal models are also altered in humans. On the epidemiological side, longitudinal cohort studies combining gold-standard urinary sodium biomarkers with serial neuroimaging, fluid biomarkers of neurodegeneration, and where possible post-mortem neuropathological end points are essential. Sex-stratified analyses, attention to mid-life exposure windows, and studies designed to span both the high and low ends of the sodium intake distribution should be design priorities. Racial and ethnic differences in the cerebrovascular and cognitive response to dietary sodium also remain unexplored and warrant dedicated investigation, particularly given known racial disparities

in salt sensitivity of BP. Finally, adequately powered randomized controlled trials with cognitive end points and sufficiently long follow-up are needed to establish whether dietary sodium reduction can prevent or slow cognitive decline, and to identify the intake range most protective for long-term brain health. Until such trials are conducted, the current evidence base, while not yet sufficient to establish causality, supports a biologically plausible and clinically relevant link between excessive sodium intake and cognitive impairment, consistent with the broader public health rationale for sodium reduction.

ARTICLE INFORMATION

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Disclosures

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REFERENCES

1. Faraco G. Dietary salt, vascular dysfunction, and cognitive impairment. *Cardiovasc Res*. 2025;120:2349–2359. doi: 10.1093/cvr/cvae229
2. Liu Y, Rusch NJ, Lombard JH. Loss of endothelium and receptor-mediated dilation in pial arterioles of rats fed a short-term high salt diet. *Hypertension*. 1999;33:686–688. doi: 10.1161/01.hyp.33.2.686
3. Lombard JH, Sylvester FA, Phillips SA, Frisbee JC. High-salt diet impairs vascular relaxation mechanisms in rat middle cerebral arteries. *Am J Physiol Heart Circ Physiol*. 2003;284:H1124–H1133. doi: 10.1152/ajpheart.00835.2002
4. Sylvester FA, Stepp DW, Frisbee JC, Lombard JH. High-salt diet depresses acetylcholine reactivity proximal to NOS activation in cerebral arteries. *Am J Physiol Heart Circ Physiol*. 2002;283:H353–H363. doi: 10.1152/ajpheart.00127.2002
5. Foulquier S, Dupuis F, Perrin-Sarrado C, Maguin Gate K, Merhi-Soussi F, Liminana P, Kwan YW, Capdeville-Atkinson C, Lartaud I, Atkinson J. High salt intake abolishes AT(2)-mediated vasodilation of pial arterioles in rats. *J Hypertens*. 2011;29:1392–1399. doi: 10.1097/HJH.0b013e328347050e
6. Durand MJ, Lombard JH. Low-dose angiotensin II infusion restores vascular function in cerebral arteries of high salt-fed rats by increasing copper/zinc superoxide dimutase expression. *Am J Hypertens*. 2013;26:739–747. doi: 10.1093/ajh/hpt015
7. Faraco G, Brea D, Garcia-Bonilla L, Wang G, Racchumi G, Chang H, Buendia I, Santisteban MM, Segarra SG, Koizumi K, et al. Dietary salt promotes neurovascular and cognitive dysfunction through a gut-initiated TH17 response. *Nat Neurosci*. 2018;21:240–249. doi: 10.1038/s41593-017-0059-z
8. Faraco G, Hochrainer K, Segarra SG, Schaeffer S, Santisteban MM, Menon A, Jiang H, Holtzman DM, Anrather J, Iadecola C. Dietary salt promotes cognitive impairment through tau phosphorylation. *Nature*. 2019;574:686–690. doi: 10.1038/s41586-019-1688-z
9. Lindauer U, Villringer A, Dirnagl U. Characterization of CBF response to somatosensory stimulation: model and influence of anesthetics. *Am J Physiol*. 1993;264:H1223–H1228. doi: 10.1152/ajpheart.1993.264.4.H1223
10. Niwa K, Haensel C, Ross ME, Iadecola C. Cyclooxygenase-1 participates in selected vasodilator responses of the cerebral circulation. *Circ Res*. 2001;88:600–608. doi: 10.1161/01.res.88.6.600
11. Cosic A, Jukic I, Stupin A, Mihalj M, Mihaljevic Z, Novak S, Vukovic R, Drenjancevic I. Attenuated flow-induced dilatation of middle cerebral arteries

is related to increased vascular oxidative stress in rats on a short-term high salt diet. *J Physiol*. 2016;594:4917–4931. doi: 10.1113/JP272297

12. McEwen ST, Schmidt JR, Somborg L, Cruz L, Lombard JH. Time-course and mechanisms of restored vascular relaxation by reduced salt intake and angiotensin II infusion in rats fed a high-salt diet. *Microcirculation*. 2009;16:220–234. doi: 10.1080/10739680802544177
13. Allen LA, Schmidt JR, Thompson CT, Carlson BE, Beard DA, Lombard JH. High salt diet impairs cerebral blood flow regulation via salt-induced angiotensin II suppression. *Microcirculation*. 2019;26:e12518. doi: 10.1111/micc.12518
14. Taheri S, Yu J, Zhu H, Kindy MS. High-sodium diet has opposing effects on mean arterial blood pressure and cerebral perfusion in a transgenic mouse model of Alzheimer's disease. *J Alzheimers Dis*. 2016;54:1061–1072. doi: 10.3233/JAD-160331
15. Yu J, Zhu H, Kindy MS, Taheri S. The impact of a high-sodium diet regimen on cerebrovascular morphology and cerebral perfusion in Alzheimer's disease. *Cereb Circ Cogn Behav*. 2023;4:100161. doi: 10.1016/j.cccb.2023.100161
16. Kleiweietfeld M, Manzel A, Titze J, Kvakan H, Yosef N, Linker RA, Muller DN, Hafler DA. Sodium chloride drives autoimmune disease by the induction of pathogenic TH17 cells. *Nature*. 2013;496:518–522. doi: 10.1038/nature11868
17. Wu C, Yosef N, Thalhammer T, Zhu C, Xiao S, Kishi Y, Regev A, Kuchroo VK. Induction of pathogenic TH17 cells by inducible salt-sensing kinase SGK1. *Nature*. 2013;496:513–517. doi: 10.1038/nature11984
18. Santisteban MM, Schaeffer S, Anfray A, Faraco G, Brea D, Wang G, Sobanko MJ, Sciortino R, Racchumi G, Waisman A, et al. Meningeal interleukin-17-producing T cells mediate cognitive impairment in a mouse model of salt-sensitive hypertension. *Nat Neurosci*. 2024;27:63–77. doi: 10.1038/s41593-023-01497-z
19. Ahn SJ, Goya B, Bertomo C, Sciortino R, Racchumi G, Bonilla LG, Anrather J, Iadecola C, Faraco G. Neutrophil stalling does not mediate the increase in tau phosphorylation and the cognitive impairment associated with high salt diet. *Cardiovasc Res*. 2025;121:2578–2593. doi: 10.1093/cvr/cvaf217
20. Santisakultarm TP, Paduano CQ, Stokol T, Southard TL, Nishimura N, Skoda RC, Olbricht WL, Schafer AL, Silver RT, Schaffer CB. Stalled cerebral capillary blood flow in mouse models of essential thrombocythemia and polycythemia vera revealed by in vivo two-photon imaging. *J Thromb Haemost*. 2014;12:2120–2130. doi: 10.1111/jth.12738
21. Cruz Hernandez JC, Bracko O, Kersbergen CJ, Muse V, Haft-Javaherian M, Berg M, Park L, Vinarsik LK, Ivasyk I, Rivera DA, et al. Neutrophil adhesion in brain capillaries reduces cortical blood flow and impairs memory function in Alzheimer's disease mouse models. *Nat Neurosci*. 2019;22:413–420. doi: 10.1038/s41593-018-0329-4
22. Bracko O, Vinarsik LK, Cruz Hernandez JC, Ruiz-Urbe NE, Haft-Javaherian M, Falkenhain K, Ramanauskaitė EM, Ali M, Mohapatra A, Swallow MA, et al. High fat diet worsens Alzheimer's disease-related behavioral abnormalities and neuropathology in APP/PS1 mice, but not by synergistically decreasing cerebral blood flow. *Sci Rep*. 2020;10:9884. doi: 10.1038/s41598-020-65908-y
23. El Amki M, Gluck C, Binder N, Middleham W, Wyss MT, Weiss T, Meister H, Luft A, Weller M, Weber B, et al. Neutrophils obstructing brain capillaries are a major cause of no-reflow in ischemic stroke. *Cell Rep*. 2020;33:108260. doi: 10.1016/j.celrep.2020.108260
24. Rolfes L, Riek-Burchardt M, Pawlitzki M, Minnerup J, Bock S, Schmidt M, Meuth SG, Gunzer M, Neumann J. Neutrophil granulocytes promote flow stagnation due to dynamic capillary stalls following experimental stroke. *Brain Behav Immun*. 2021;93:322–330. doi: 10.1016/j.bbi.2021.01.011
25. Erdener SE, Tang J, Kilic K, Postnov D, Giblin JT, Kura S, Chen IA, Vayisoglu T, Sakadzic S, Schaffer CB, et al. Dynamic capillary stalls in reperfused ischemic penumbra contribute to injury: a hyperacute role for neutrophils in persistent traffic jams. *J Cereb Blood Flow Metab*. 2021;41:236–252. doi: 10.1177/0271678X20914179
26. Ali M, Falkenhain K, Njiru BN, Murtaza-Ali M, Ruiz-Urbe NE, Haft-Javaherian M, Catchers S, Nishimura N, Schaffer CB, Bracko O. VEGF signalling causes stalls in brain capillaries and reduces cerebral blood flow in Alzheimer's mice. *Brain*. 2022;145:1449–1463. doi: 10.1093/brain/awab387
27. Sharma S, Cheema M, Reeson PL, Narayana K, Boghoozian R, Cota AP, Brosschot TP, FitzPatrick RD, Korbelin J, Reynolds LA, et al. A pathogenic role for IL-10 signalling in capillary stalling and cognitive impairment in type 1 diabetes. *Nat Metab*. 2024;6:2082–2099. doi: 10.1038/s42255-024-01159-9
28. Lim HK, Bae S, Han K, Kang BM, Jeong Y, Kim SG, Suh M. Seizure-induced neutrophil adhesion in brain capillaries leads to a decrease in postictal cerebral blood flow. *iScience*. 2023;26:106655. doi: 10.1016/j.isci.2023.106655
29. Yuan M, Wang Y, Wen J, Jing F, Zou Q, Pu Y, Pan T, Cai Z. Dietary salt disrupts tricarboxylic acid cycle and induces tau hyperphosphorylation and synapse dysfunction during aging. *Aging Dis*. 2022;13:1532–1545. doi: 10.14336/AD.2022.0220
30. Sun F, Wang QX, Zhao LP, Jin Q, Jin SH, Xu JT, Yin MJ, Jin C, Wang JH. Sex-specific effects of dietary salt intake on circulating Alzheimer's disease-related biomarkers in aged rats. *J Alzheimers Dis Rep*. 2025;9:25424823251315388. doi: 10.1177/25424823251315388
31. Rui TY, Huang HZ, Zheng K, Fan HW, Zhang J, Guo ZY, Man HY, Brazhe N, Semyanov A, Lu YM, et al. Tau pathology drives disease-associated astrocyte reactivity in salt-induced neurodegeneration. *Adv Sci (Weinh)*. 2025;12:e2410799. doi: 10.1002/adv.202410799
32. Gu N, Makashova O, Laporte C, Chen CQ, Li B, Chevillard PM, Lean G, Yang J, Wong C, Fan J, et al. Microglia regulate neuronal activity via structural remodeling of astrocytes. *Neuron*. 2025;113:3408–3423.e5. doi: 10.1016/j.neuron.2025.07.024
33. Hu L, Zhu S, Peng X, Li K, Peng W, Zhong Y, Kang C, Cao X, Liu Z, Zhao B. High salt elicits brain inflammation and cognitive dysfunction, accompanied by alternations in the gut microbiota and decreased SCFA production. *J Alzheimers Dis*. 2020;77:629–640. doi: 10.3233/JAD-200035
34. Heras-Garvin A, Refolo V, Reindl M, Wenning GK, Stefanova N. High-salt diet does not boost neuroinflammation and neurodegeneration in a model of α -synucleinopathy. *J Neuroinflammation*. 2020;17:35. doi: 10.1186/s12974-020-1714-y
35. Yang X, Liu S, Wang C, Fan H, Zou Q, Pu Y, Cai Z. Dietary salt promotes cognition impairment through GLP-1R/mTOR/p70S6K signaling pathway. *Sci Rep*. 2024;14:7970. doi: 10.1038/s41598-024-57998-9
36. Kubota H, Kunisawa K, Wulaer B, Hasegawa M, Kurahashi H, Sakata T, Tezuka H, Kugita M, Nagao S, Nagai T, et al. High salt induces cognitive impairment via the interaction of the angiotensin II-AT1 and prostaglandin E2-EP1 systems. *Br J Pharmacol*. 2023;180:2393–2411. doi: 10.1111/bph.16093
37. Chugh G, Asghar M, Patki G, Bohat R, Jafri F, Allam F, Dao AT, Mowrey C, Alkadhi K, Salim S. A high-salt diet further impairs age-associated declines in cognitive, behavioral, and cardiovascular functions in male Fischer brown Norway rats. *J Nutr*. 2013;143:1406–1413. doi: 10.3945/jn.113.177980
38. Sun F, Zhang JY, Zhao LP, Jin Q, Jin C, Wang QX, Xu JT, Yin MJ. Age differences in the impact of dietary salt on metabolism, blood pressure and cognitive function in male rats. *Food Funct*. 2024;15:689–703. doi: 10.1039/d3fo04211a
39. Liu YZ, Chen JK, Li ZP, Zhao T, Ni M, Li DJ, Jiang CL, Shen FM. High-salt diet enhances hippocampal oxidative stress and cognitive impairment in mice. *Neurobiol Learn Mem*. 2014;114:10–15. doi: 10.1016/j.nlm.2014.04.010
40. Ge Q, Wang Z, Wu Y, Huo Q, Qian Z, Tian Z, Ren W, Zhang X, Han J. High salt diet impairs memory-related synaptic plasticity via increased oxidative stress and suppressed synaptic protein expression. *Mol Nutr Food Res*. 2017;61:1700134. doi: 10.1002/mnfr.201700134
41. Wang S, Deng K, Zhao H, Wang Y, Guo P, Jin H, Qi L. Analysis of the association between dietary sodium intake and cognitive function: a NHANES-based machine learning study and animal experimental validation. *Front Nutr*. 2025;12:1626651. doi: 10.3389/fnut.2025.1626651
42. Fan H, Yuan M, Wang S, Yang X, Shu L, Pu Y, Zou Q, Zhang X, Wang C, Cai Z. Dietary salt promotes cognitive impairment through repression of SIRT3/PINK1-mediated mitophagy and fission. *Mol Cell Biochem*. 2025;480:2345–2360. doi: 10.1007/s11010-024-05069-y
43. Guo CP, Wei Z, Huang F, Qin M, Li X, Wang YM, Wang Q, Wang JZ, Liu R, Zhang B, et al. High salt induced hypertension leads to cognitive defect. *Oncotarget*. 2017;8:95780–95790. doi: 10.18632/oncotarget.21326
44. Ge Q, Hu X, Ma N, Sun M, Zhang L, Cai Z, Tan R, Lu H. Maternal high-salt diet during pregnancy impairs synaptic plasticity and memory in offspring. *FASEB J*. 2021;35:e21244. doi: 10.1096/fj.202001890R
45. Mohan D, Yap KH, Reidpath D, Soh YC, McGrattan A, Stephan BCM, Robinson L, Chaiyakunapruk N, Siervo M; DePEC team. Link between dietary sodium intake, cognitive function, and dementia risk in middle-aged and older adults: a systematic review. *J Alzheimers Dis*. 2020;76:1347–1373. doi: 10.3233/JAD-191339
46. Afsar B. The relationship between cognitive function, depressive behaviour and sleep quality with 24-h urinary sodium excretion in patients with essential hypertension. *High Blood Press Cardiovasc Prev*. 2013;20:19–24. doi: 10.1007/s40292-013-0002-7
47. Rush TM, Kritzer-Silverstein D, Laughlin GA, Fung TT, Barrett-Connor E, McEvoy LK. Association between dietary sodium intake and cognitive function in older adults. *J Nutr Health Aging*. 2017;21:276–283. doi: 10.1007/s12603-016-0766-2

48. Wang Z, Li N, Heizhati M, Wang L, Li M, Pan F, Yang Z, Abudureyimu R, Hong J, Sun L, et al. Association between 24-h urinary sodium to potassium ratio and mild cognitive impairment in community-based general population. *Public Health Nutr.* 2021;24:5795–5804. doi: 10.1017/S1368980021001452
49. McGrattan AM, Stephan BCM, Shannon OM, Mazidi M, Gilchrist M, Smallwood M, Winyard P, McMahon N, Blekkenhorst LC, Mohan D, et al. Independent and interactive associations of dietary nitrate and salt intake with blood pressure and cognitive function: a cross-sectional analysis in the InCHIANTI study. *Int J Food Sci Nutr.* 2022;73:491–502. doi: 10.1080/09637486.2021.1993157
50. Fiocco AJ, Shatenstein B, Ferland G, Payette H, Belleville S, Kergoat MJ, Morais JA, Greenwood CE. Sodium intake and physical activity impact cognitive maintenance in older adults: the NuAge Study. *Neurobiol Aging.* 2012;33:829.e21–829.e28. doi: 10.1016/j.neurobiolaging.2011.07.004
51. Haring B, Wu C, Coker LH, Seth A, Snetselaar L, Manson JE, Rossouw JE, Wassertheil-Smoller S. Hypertension, dietary sodium, and cognitive decline: results from the women's health initiative memory study. *Am J Hypertens.* 2016;29:202–216. doi: 10.1093/ajh/hpv081
52. Nowak KL, Fried L, Jovanovich A, Ix J, Yaffe K, You Z, Chonchol M. Dietary sodium/potassium intake does not affect cognitive function or brain imaging indices. *Am J Nephrol.* 2018;47:57–65. doi: 10.1159/000486580
53. Milte CM, Ball K, Crawford D, McNaughton SA. Diet quality and cognitive function in mid-aged and older men and women. *BMC Geriatr.* 2019;19:361. doi: 10.1186/s12877-019-1326-5
54. Klinedinst BS, Le ST, Larsen B, Pappas C, Hoth NJ, Pollpeter A, Wang Q, Wang Y, Yu S, Wang L, et al. Genetic factors of Alzheimer's disease modulate how diet is associated with long-term cognitive trajectories: a UK biobank study. *J Alzheimers Dis.* 2020;78:1245–1257. doi: 10.3233/JAD-201058
55. Na X, Xi M, Zhou Y, Yang J, Zhang J, Xi Y, Yang Y, Yang H, Zhao A. Association of dietary sodium, potassium, sodium/potassium, and salt with objective and subjective cognitive function among the elderly in China: a prospective cohort study. *Global Transitions.* 2022;4:28–39. doi: 10.1016/j.glt.2022.10.002
56. Liu W, Xing S, Wei F, Yao Y, Zhang H, Li YC, Liu Z. Excessive dietary salt intake exacerbates cognitive impairment progression and increases dementia risk in older adults. *J Am Med Dir Assoc.* 2023;24:125–129.e4. doi: 10.1016/j.jamda.2022.10.001
57. Liu D, Zhang Q, Xing S, Wei F, Li K, Zhao Y, Zhang H, Gong G, Guo Y, Liu Z. Excessive salt intake accelerates the progression of cerebral small vessel disease in older adults. *BMC Geriatr.* 2023;23:263. doi: 10.1186/s12877-023-03877-3
58. Makin SDJ, Mubki GF, Doubal FN, Shuler K, Staals J, Dennis MS, Wardlaw JM. Small vessel disease and dietary salt intake: cross-sectional study and systematic review. *J Stroke Cerebrovasc Dis.* 2017;26:3020–3028. doi: 10.1016/j.jstrokecerebrovasdis.2017.08.004
59. Ahn SH, Jeong JH, Park KW, Kim EJ, Yoon SJ, Yoon B, Jang JW, Minn Y, Choi SH. Effect of dietary habits on Alzheimer's disease progression. *Yonsei Med J.* 2024;65:217–226. doi: 10.3349/ymj.2023.0119
60. Tong H, Capuano AW, Agarwal P, Mehta RI, Sood A, Bennett DA, Arvanitakis Z. Association of dietary sodium intake with late-life cognitive decline and postmortem neuropathology in community-dwelling older adults. *Alzheimers Dement.* 2025;21:e70706. doi: 10.1002/alz.70706
61. Chuwa F, Rainey-Smith SR, Al Shamsi HSS, Sohrabi HR, Taddei K, Masters CL, Martins RN, Gardener SL; AIBL Research Group. Higher sodium intake is associated with episodic memory decline in cognitively unimpaired older males: a 6-year longitudinal study. *Neurobiol Aging.* 2026;162:22–29. doi: 10.1016/j.neurobiolaging.2026.02.003
62. Abe C, Imai T, Sezaki A, Miyamoto K, Kawase F, Shirai Y, Sanada M, Inden A, Sugihara N, Honda T, et al. Global associations of sodium intake with dementia prevalence from 2010 to 2021 by sex: a longitudinal ecological study. *JAND Global Reports.* 2026;3:100015. doi: 10.1016/j.jaglob.2026.100015
63. van der Burgh AC, Pelouto A, Mooldijk SS, Zandbergen AAM, Ikram MA, Chaker L, Hoorn EJ. Serum sodium, cognition and incident dementia in the general population. *Age Ageing.* 2023;52:afad007. doi: 10.1093/ageing/afad007
64. Lee S, Min JY, Kim B, Ha SW, Han JH, Min KB. Serum sodium in relation to various domains of cognitive function in the elderly US population. *BMC Geriatr.* 2021;21:328. doi: 10.1186/s12877-021-02260-4
65. Hui Z, Wang L, Deng J, Liu F, Cheng L, Li Y, Tian Y, Ma L, Liu X. Joint association of serum sodium and frailty with mild cognitive impairment among hospitalized older adults with chronic diseases: a cross-sectional study. *Front Nutr.* 2024;11:1467751. doi: 10.3389/fnut.2024.1467751
66. Nowak KL, Yaffe K, Orwoll ES, Ix JH, You Z, Barrett-Connor E, Hoffman AR, Chonchol M. Serum sodium and cognition in older community-dwelling men. *Clin J Am Soc Nephrol.* 2018;13:366–374. doi: 10.2215/CJN.07400717
67. Chen YH, Wang ZB, Liu XP, Mao ZQ; Alzheimer's Disease Neuroimaging Initiative. Elevated serum sodium is linked to increased amyloid-dependent tau pathology, neurodegeneration, and cognitive impairment in Alzheimer's disease. *J Neurochem.* 2025;169:e16257. doi: 10.1111/jnc.16257
68. Blumenthal JA, Smith PJ, Mabe S, Hinderliter A, Lin PH, Liao L, Welsh-Bohmer KA, Browndyke JN, Kraus WE, Doraiswamy PM, et al. Lifestyle and neurocognition in older adults with cognitive impairments: a randomized trial. *Neurology.* 2019;92:e212–e223. doi: 10.1212/WNL.0000000000006784
69. Salerno-Kennedy R, Cashman KD. The relationship between nutrient intake and cognitive performance in people at risk of dementia. *Ir J Med Sci.* 2007;176:193–198. doi: 10.1007/s11845-007-0036-8
70. Meneton P, Jeunemaitre X, de Wardener HE, MacGregor GA. Links between dietary salt intake, renal salt handling, blood pressure, and cardiovascular diseases. *Physiol Rev.* 2005;85:679–715. doi: 10.1152/physrev.00056.2003
71. Dmitrieva NI, Gagarin A, Liu D, Wu CO, Boehm M. Middle-age high normal serum sodium as a risk factor for accelerated biological aging, chronic diseases, and premature mortality. *EBioMedicine.* 2023;87:104404. doi: 10.1016/j.ebiom.2022.104404